

CV Research-Reproduction Project

Your mission: Pick a healthcare CV topic, find a research paper you can reproduce, reproduce its result, then propose and test ONE change of your own - and write it up like a real conference paper. Your marks come from your predictions, experiments, and defence.

The six steps

#	Step	What you do
1	Literature survey	Read 10-20 recent papers on your topic; note methods, datasets, metrics.
2	Pick a base paper	Choose ONE (possibly) with public code + a downloadable dataset. Fill the reproducibility checklist.
3	Reproduce	Re-run its method on the same data; land near its reported metric (exact match not required).
4	Your hypothesis + Introduction	Propose ONE change + predict the effect and why - submit BEFORE you code.
5	Implement & test	Make the one change; run ablations; measure what moved.
6	Analyse & write	Failure analysis + metrics; write it up in the IEEE paper template; defend it.

Topics & datasets (pick one)

Topic	Dataset	Task
Diabetic retinopathy grading	APTOS 2019	Classification (transfer learning)
Retinal OCT disease	Kermany OCT	Classification
Pneumonia (chest X-ray)	Kermany CXR	Classification
Skin lesion / melanoma	HAM10000	Classification
Malaria cell detection	NIH Malaria	Classification
Brain tumor (MRI)	Brain Tumor MRI	Classification
Polyp segmentation	Kvasir-SEG	Segmentation (U-Net)
Retinal vessel segmentation	DRIVE	Segmentation (U-Net)

Submit at each milestone

- **M1:** survey + chosen paper + reproducibility checklist • **M2:** baseline reproduced (it runs)
- **M3:** hypothesis + prediction (before coding) + Introduction • **M4:** implementation + ablation results
- **M5:** failure analysis + IEEE-format paper + defence

Two rules: (1) Predict before you experiment - being wrong is fine, good reasoning is what counts. (2) No clinical claims; report sensitivity/recall + AUC, not just accuracy, and include a limitations section.